

A machine learning approach to analysis of Canadian provincial power system decarbonization

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ABSTRACT

Canada committed to reducing national greenhouse gas emissions by 40–45 % by 2030 and achieving net-zero by 2050, relying on low-carbon electrification across economic sectors. This study investigates the machine learning's potential to analyze least-cost optimization of Canadian provincial power systems to 2050, to identify robust pathways for net-zero electricity system greenhouse gas emissions, and to provide policy insights respecting wind and solar generation, reducing emissions from natural gas generation, and expanding inter-provincial transmission. Surrogate modeling is applied as sensitivities to generation technology capital costs, electricity demand, and carbon pricing to results from an existing capacity expansion mixed-integer linear optimization model. Under almost all future scenarios, six provinces develop new wind generation, with substantial deployment in Alberta (mean 27.3 GW, max. 62.7 GW), Ontario (mean 6.8 GW, max. 19.3 GW), Saskatchewan (mean 3.6 GW, max 8.9 GW) and Manitoba (mean 1.5 GW, max 3.5 GW). With the potential exception of BC, where higher-cost wind generation results in solar generation becoming the primary low-carbon electrical energy source (mean 25.2 GW), future solar generation across Canada is more limited (mean 37.1 GW) and less certain than wind generation in all other Canadian provinces. Existing natural gas generation is maintained in almost all scenarios in all provinces, providing support for maintaining these resources for reliability and reserve requirements. Very limited natural gas is developed under any scenarios, with the exception of British Columbia (mean 4.3 GW) where more than 80 % of scenarios include new natural gas capacity. Additional transmission transfer capacity is developed in most scenarios between BC-Alberta (~60 % of scenarios), Saskatchewan-Manitoba (~85 %), and Ontario-Quebec (~75 %). Carbon prices, along with wind and solar capital costs, act as primary determinants of least-cost decarbonization pathways. These findings offer insights for Canadian policymakers into net-zero pathways for provincial electricity systems that remain robust despite variability in inputs.

1. Introduction and motivation

As a party to the Paris Agreement, Canada committed to reducing national economy-wide greenhouse gas emissions (GHG) by 40–45 % by 2030 (Government of Canada, 2023a) and to net-zero by 2050 (Canada Net-Zero Emissions Accountability Act, 2021). Achievement depends upon simultaneous end-use electrification, electricity generation decarbonization, and electrical technology evolution (Vaillancourt et al., 2017; Arjmand and McPherson, 2022), necessitating expansion and decarbonization of the power sector. Decarbonization analyses of

Canada's electricity sector considering uncertainties in technology costs, electricity demand and effects of carbon pricing result in very large buildouts of hydroelectric, wind or nuclear generation (Arjmand and McPherson, 2022; The Canadian Academy of Engineering, 2016; Bataille et al., 2023) or significant transmission expansions (Dolter and Fellows, 2022). These studies focus on determining technically feasible or economically viable national strategies, an approach with disadvantages for determining robust pathways to decarbonization considering policy uncertainties and model limitations.

Since electricity services in Canada are provincially regulated,

Abbreviations: AB, Alberta; BC, British Columbia; MB, Manitoba; NB, New Brunswick; NS, Nova Scotia; NL, Newfoundland and Labrador; ON, Ontario; PE, Prince Edward Island; QC, Quebec; SK, Saskatchewan.

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achieving national emissions reduction targets must address differing provincial economies (CER – Provincial and Territorial Energy Profiles – Canada, 2023), geographies, technology costs, and policy realities (Carbon pollution pricing systems across Canada - Canada.ca, 2023). Canadian provincial economies differ substantially in their sectoral contributions to gross domestic product (GDP) and in their per capita GDP (Fig. 1). Contributions of the energy sector (i.e., oil, natural gas and coal production and distribution) to GDP range from a fraction (1.2 % in PE) to nearly a quarter (23.0 % in AB) of provincial economies, while contributing proportionately to provincial emissions and substantially (28.2 %) to national emissions (Greenhouse gas emissions - Canada.ca, 2024). The range in per capita GDP (~\$42k in the Maritimes to ~72k in Alberta) reflects differences in consumer ability to finance the costs of electricity decarbonization (IESO, 2023a; AESO, 2022). Some Canadian provinces have abundant variable renewable, hydroelectric, or nuclear generation while others rely heavily on fossil fuels, resulting in electricity sector GHG emissions intensities ranging from 1 to 670 tCO₂e/GWh (Government of Canada, 2023a) (Fig. 2). Others face geographical constraints on development, lack access to low-carbon resources capable of integrating variable renewables (e.g., storage hydroelectric), lack high-quality geological storage sites, or require transmission expansion to integrate remote resources, all of which contribute to variability in the costs of electricity decarbonization (Diner et al., 2023).

Notwithstanding provincial regulation of the electricity sector, Canadian provinces are subject to federal regulations supporting national decarbonization efforts (Government of Canada, 2024a). These include economy-wide carbon pricing with an output-based pricing system (OBPS) to mitigate adverse effects on competitiveness (Greenhouse Gas Pollution Pricing Act, 2018a; Output-Based Pricing System Regulations; Government of Canada, 2024b), phasing out conventional coal-fired generation by 2030 (Reduction of Carbon Dioxide Emissions.), and imposing performance standards (370 tCO₂e/GWh) on natural gas-fired generation (Regulations Limiting Carbon Dioxide Emissions.). In addition, to limit remaining emissions from natural gas generation beyond 2030 (Arjmand and McPherson, 2022), pending regulations require net-zero provincial electricity systems by 2035 (Government of Canada, 2022).

Canadian provinces implementing fossil-fuel generation phase-out, renewable portfolio standards, and feed-in-tariffs successfully reduce electricity sector emissions though, misalignment of federal and provincial policies could limit national emission reductions (Jordaan et al., 2017). For example, some but not all provinces prohibit nuclear (Nova Scotia Power Privatization Act, 1992) or large hydroelectric development (Clean Energy Act, 2010), renewable energy and GHG reduction

targets differ across provinces, while provincial carbon pricing regimes also vary despite the federal carbon pricing mechanism (Carbon pollution pricing systems across Canada - Canada, 2023). Recent studies also indicate that least-cost decarbonization in some provinces entails development of inter-provincial (Dolter and Rivers, 2018; Brinkman et al., 2024) and international transmission (Brinkman et al., 2024; Rodríguez-Sarasty et al., 2021). Multi-year federal efforts aimed at securing substantially enhanced regional transmission in Atlantic Canada (Natural Resources Canada, 2022) recently concluded with provincial utilities opting for a more modest expansion of an existing transmission intertie (Joint Policy Statement on Developing and Transmitting Clean, 2024). Provincial level analyses could provide planners and policymakers with in-depth understanding of the challenges and opportunities specific to each province, while producing deeper insights into policies for developing variable renewable generation, reducing emissions from natural gas generation, and expanding interprovincial transmission.

The historic focus on pathway analysis also lends itself to electricity system modeling centered on least-cost optimization, simulation, and single-factor sensitivity. This conventional approach has limitations relevant to evaluating net-zero pathway robustness, which requires comprehensive examination of strategies and technologies in the context of uncertainties inherent to evolving and interconnected power systems. First, conventional approaches are computationally intensive, limiting available scenarios for exploring sensitivities associated with alternative futures. Secondly, these approaches typically assume deterministic relationships between variable inputs and model outputs, limiting their application for decision-making under uncertainty. Power systems are subject to significant variability in technology costs, operations, policy constraints, and future electricity requirements, particularly on the multi-decadal horizon of deep decarbonization. Thirdly, conventional models often exclude correlations between model parameters, and between parameters and input variables. In policy-setting and decision-making, such correlations are often salient to determining opportunities for implementing deeper, more cost-effective, or less risky decarbonization pathways.

Machine learning (ML) models offer a potential solution to charting decarbonization of Canadian provincial electricity systems while accounting for uncertainties and variabilities. A recent review of ML techniques demonstrated increased interest in using ML to address technical challenges in evolving power systems (Ibrahim et al., 2020a). These challenges in forecasting renewable energy generation and future deployment stem from multifaceted uncertainties inherent to wind (Nazir et al., 2020) and solar (Alhamer et al., 2022) resources and to evolution of technologies (US Department of Energy Wind Energy

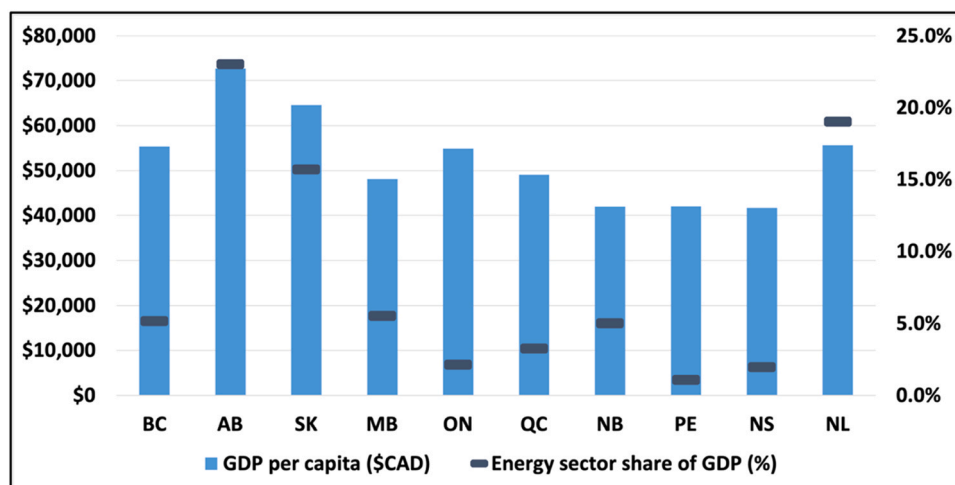


Fig. 1. Canadian provincial differences in per capita GDP and energy sector share of GDP (Statistics Canada, 2024a, 2024b).

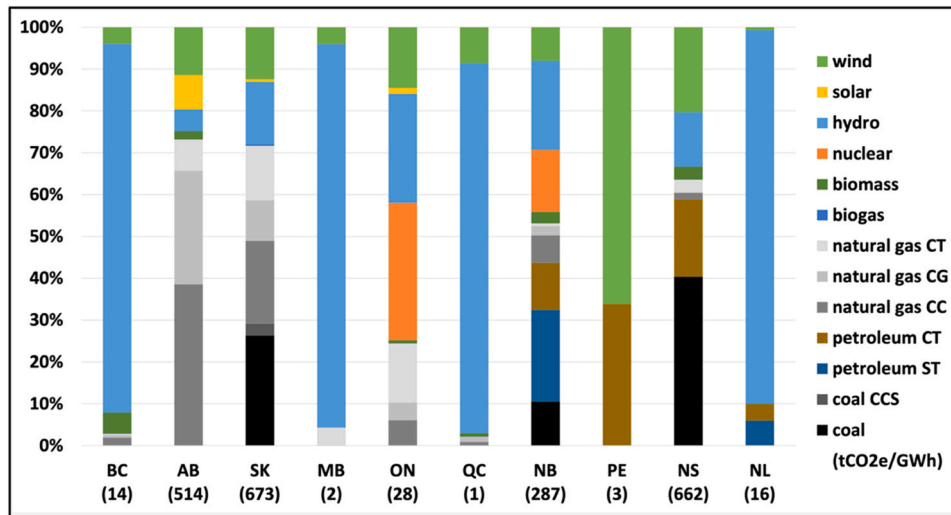


Fig. 2. Canadian electricity generation mix and emissions intensities by province.

Technologies Office, 2023) and costs (Ramamamy et al., 2022). These uncertainties involve complex non-linear relationships between input parameters and outcomes, and the integration of real-world uncertainties into modeling frameworks. Estimating rare events and extreme outcomes poses additional challenges due to their infrequency and impact. Moreover, long-term electricity system planning horizons and interlinked socio-technical factors exacerbate uncertainty, particularly in detailed modeling of optimization strategies.

By capturing complex correlations and relationships between model parameters and input variables, ML models account for uncertainty and variability in wind and solar generation (Benti et al., 2023) and power system evolution (Javed et al., 2022), supporting more robust decarbonization strategies that are optimal or near optimal under a wide variety of future conditions. In the context of the large datasets inherent to multi-factor uncertainty and variability, ML models can also be trained to learn patterns and relationships not apparent to conventional models (Ibrahim et al., 2020b). An exploration of ML techniques in power systems highlighted advantages and limitations compared to conventional techniques and concluded that ML offers significant potential for real-time applications in power systems (Miraftabzadeh et al., 2019). ML methods have been applied to load forecasting and power system optimization problems (Asgharnejad Keisami et al., 2021), to determining the importance of stakeholder and planning criteria in system expansion planning (Majid Oloomi Buygi, Hasan Modir Shanechi et al., 2006), and to solving complex control problems in power systems (Zidong Zhang et al., 2020). A recent analysis utilized ML to integrate principal component analysis and clustering techniques to determine representative days for a reduced capacity expansion model applied to model grid decarbonization (Zhao and You, 2022).

These investigations demonstrate the potential of ML models to improve the accuracy and efficiency of power system planning and modeling, enabling more informed and effective decision-making. However, none of these studies explore the potential of ML in capacity expansion models to explore correlations, uncertainties, and variability in different input parameters affecting power system evolution. Exploiting the capabilities of ML methods in capacity expansion planning could improve policymaking and planning for decarbonizing these regional electricity systems (Jahangiri et al., 2023a).

Contributions. Our contributions to energy systems modeling using ML models are as follows:

- Analyzing robust least-cost pathways to net-zero electricity system decarbonization in Canada considering technology and policy uncertainties.

- Investigating the potential of ML models to analyze provincial power systems for net-zero emissions in response to capital cost, demand and carbon tax uncertainties and variability.
- Utilizing ML for sensitivity and uncertainty analysis in provincial power system variables, reducing computational costs.

2. Methodology

2.1. Model development

An ML surrogate methodology is applied to COPPER, a capacity expansion mixed-integer linear planning model, to achieve a more efficient and comprehensive analysis of output variabilities. COPPER determines the least-cost mix of generation and transmission capacity for a specific time period while adhering to power system and policy constraints. The objective function minimizes total system costs including investment, maintenance, production, and carbon costs over the study period (Eq. 1).

$$\text{Mintotal_system_costs} = \min(\text{InvestmentCost} + \text{MaintenanceCost} + \text{ProductionCost} + \text{CarbonCost}) \quad (1)$$

COPPER encompasses various optimization constraints, including those related to power balance, planning reserves, thermal generation, hydroelectric development, renewable generation, transmission expansion, and energy storage. Inputs include forecasted electricity demand, existing and future transmission, existing and future generation, and federal and provincial environmental and decarbonization policies. Though valuable for evaluating the effects of carbon pricing and decarbonization policies on grid evolution, COPPER is computationally intensive, which limits its practicality for evaluating outputs under future uncertainties (Jahangiri et al., 2023a).

The current study investigates key Canadian provincial electricity system decarbonization outcomes for 2050, encompassing carbon emissions, power generation technologies and capacities, and inter-provincial transmission capacity. These outcomes offer a comprehensive understanding of power system evolution under select input parameter variations, including carbon prices, demand growth, and generation technology costs (Table 1). Ranges for input variables are based on existing legislation, consultations with subject matter experts, and extension of upper and lower bounds by ~25 %, as is common practice in sensitivity analysis (Jahangiri et al., 2023a). For example, existing carbon pricing legislation increases prices by \$15/t/year from \$65/t in 2023 to \$170/t in 2030 but is unregulated beyond that date (Greenhouse Gas Pollution Pricing Act, 2018b). The lower bound envisions

Table 1
Specified ranges for input variables.

Input variable	Range	Unit	Rationale and References
Carbon tax	95 – 645	\$/t	\$65 in 2023, increasing annually by \$0 to \$30/t to 2030, and thereafter by \$0 to \$20/t to 2050 (Jahangiri et al., 2023a, 2023b; Greenhouse Gas Pollution Pricing Act, 2018b)
Total demand growth 2050/2018	1.2 – 2.5	ratio	Canada’s Energy Future Report (Canada Energy Regulator, 2023a) for 2050, with 2018 as the base year
Annualized capital costs of gas combined cycle	106 – 198	\$/kW	Interquartile range (IQR) determined across these estimates, and used to update the range limits based on the extremes of the identified IQR and the initial limits, as detailed in prior work (Jahangiri et al., 2023b)
Annualized capital costs of gas simple cycle	84 – 145	\$/kW	Six literature sources encompassing uncertainties linked to generation technology capital costs, used to transform baseline estimates into equivalent annual payments over a fixed amortization period (CAD \$/kW/year), in line with the original COPPER model formulation.
Annualized capital costs of diesel	104 – 195	\$/kW	
Annualized capital costs of coal	289 – 549	\$/kW	
Annualized capital costs of nuclear	536 – 972	\$/kW	
Annualized capital costs of biomass	285 – 574	\$/kW	
Annualized capital costs of wind	84 – 242	\$/kW	
Annualized capital costs of solar PV	86 – 216	\$/kW	
Annualized capital costs of battery storage	216 – 463	\$/kW	

prices rising to \$95/t by 2025 (COPPER’s base year for carbon pricing) and then remaining unchanged, while the upper bound envisions prices rising to \$95/t by 2025, accelerating by \$30/t/year to \$245/t by 2030, and then incrementally increasing at \$20/t/year to \$645/t in 2050.

2.2. Model training

The COPPER model is employed using the input variables in the specified ranges (Table 1). No constraints are imposed on transmission expansion between provinces, allowing a wide array of potential transmission expansion scenarios. Additionally, no target is imposed on emissions reductions, which decline in response to increasing carbon prices and policy constraints, evolving towards net-zero but not to zero emissions by 2050. The produced model output data are cleaned to remove outliers, and normalized to a common scale to enable fair comparisons without range distortion or information loss (Eq. 2). Outputs having a variance of less than 0.001 are assumed to be constant (model noise) and ignored in the training process, ensuring a stable and trustworthy base for future analysis and interpretations.

For i in X :

$$\hat{x}_i = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)} \tag{2}$$

The neural network is trained using a dataset of COPPER input and output samples. An empirical model development approach is employed to identify optimal hyperparameters for the model architecture. The selected architecture contains one linear and eight residual blocks, each consisting of two linear layers, two batch normalizations, two dropouts and two ReLu activation functions, with 128 neurons per layer. Batch size, learning rate, and dropout ratio values are 128, 0.001, and 0.5, respectively (Jahangiri et al., 2023a).

The average R-squared value across the entire test dataset is reported in Table 2, alongside training and test errors, and indicates a high goodness of fit of model predictions to test data. This approach ensures a comprehensive assessment of the model’s overall performance.

In Table 3, the model’s run-time and resource use are compared

Table 2
Training and Test MSE losses of the model and its accuracy in predictions.

Training loss	Test loss	Mean (R ²)	Std (R ²)
0.0078	0.08	0.94	0.03

Table 3
Comparison of COPPER and surrogate model computational costs and run-time.

Model	CPU	Memory	Run-time
COPPER	16–32	64–128 GB	15 h - 72 h
Residual neural network	1	NA*	12 ms

* Except for saving the model

against COPPER, demonstrating a substantial decrease in computational cost and resource requirements. This efficiency enables more feasible scenario exploration by allowing for as many runs as needed.

2.3. Output representation

Once the ML model is trained, a sensitivity analysis reveals patterns of evolution in each provincial electricity system to 2050. All inputs are varied within the selected ranges (Table 1) using Latin Hyperbolic Sampling, resulting in over 2000 scenarios. The surrogate model outputs consist of deployments or retirements of generation (in MW) within each province, or expansions in interprovincial transmission transfer capacity (in MW). The impacts of system parameters on the future development of generation and transmission across all sensitivity scenarios are plotted in histograms specific to each technology within each province, and in maximum absolute derivative plots of standardized outputs with respect to normalized inputs. Computationally demanding, these derivative plots present normalized values demonstrating the relative impact of each input variable on the respective output value. Negative values indicate a negative correlation, positive values a positive correlation, while absolute value represents the strength of correlation. By identifying patterns, this approach provides insights into the relationships between input sensitivities and evolution of each provincial power system to 2050. These graphical approaches are further elaborated in the results presentation.

2.4. Model limitations

In conducting this study, several assumptions and limitations were considered. As a mixed-integer linear program optimization model, COPPER has several limitations relevant to interpreting the results of this analysis (Arjmand and McPherson, 2022). While the methodology employed here imposes specified ranges on input variables, biases may persist due to rapidly changing circumstances related to decarbonization assumptions, electricity demand growth, capital costs, and technology evolution. To reduce computational burdens, COPPER also makes simplifying assumptions including treating, transmission and most generation technologies as continuous rather than binary integer variables. A more realistic characterization of generation and transmission development entails modeling contemplated projects with known capacities. This might best be achieved by a regional analysis of a subset of provinces in order to diminish computational burdens. COPPER also does not account for landscape characteristics that could preclude wind and solar development and may overstate development potential in some regions. COPPER also has a limited number of balancing areas, and for most provinces does not consider intraprovincial transmission expansion, which can be important to determining optimized decarbonization pathways. Finally, imports and exports are modeled exogenously using historical data since future conditions are unknown, so significant changes to trade patterns would influence optimal decarbonization pathways. Future modelling could consider varying

electricity trade patterns to provide policy insights into domestic decarbonization pathways.

Despite its advantages, the surrogate ML model approach also has limitations in addition to those associated with COPPER. For dataset generation, COPPER's default constraints were applied to represent practical and non-contentious pathways. One key limitation is the potential loss of granularity and accuracy due to simplified representations that do not fully capture nuances in the COPPER formulation. Secondly, the reliability of the surrogate model depends upon the quality and comprehensiveness of the training data, so extrapolating the model's predictions beyond the training data could lead to unreliable results. Additionally, this ML model exhibits less sensitivity to parameters with relatively small nominal values when normalized, potentially leading to lower predictive accuracy for small-valued outputs in some provinces, which are not the focus of this analysis. Furthermore, as COPPER evolves and incorporates new features, this ML surrogate model will require retraining to maintain accuracy. Lastly, while the surrogate model provides predictive power, understanding the specific reasons behind its predictions can be challenging, which may impact its overall interpretability.

3. Results

This ML analysis produces abundant results, not all of which are meaningful. Findings are presented as national and provincial summaries in terms of additions of renewable generation (wind and solar), additions and retirements of natural gas generation, and expansions of inter-provincial transmission. Results presented are mostly non-constant, meaning they vary across different input sensitivities since such results are generally of greater interest to policymakers (Jahangiri et al., 2023a, 2023b). For example, results respecting coal generation are omitted since this technology is rapidly retired in all provinces under all scenarios due to high emissions, moderate fuel costs, and increasing carbon prices. Selection of results focuses on key provincial pairings including British Columbia (BC) – Alberta (AB), Saskatchewan (SK) – Manitoba (MB), and Ontario (ON) – Quebec (QC), which comprise more than 90 % of installed generation capacity in Canada (Canada Energy Regulator, 2023b), and where transmission interconnection expansion is under active consideration by system planners in support of grid decarbonization.

3.1. Renewable generation

Table 4 summarizes results for modeled wind and solar development

Table 4
Modeled wind and solar development by Province.

Province	WIND					SOLAR				
	Existing		Additions			Existing		Additions		
	Installed capacity		Scenarios with no additional capacity	Mean additional installed capacity	Mean additional capacity above existing	Installed capacity		Scenarios with no additional capacity	Mean additional installed capacity	Mean additional capacity above existing
	(GW)	(%)	(GW)	(%)	(GW)	(GW)	(%)	(GW)	(%)	(GW)
BC	0.7	54 %	4.3	582 %	49.4	0.0	11 %	25.2	n/a	72.2
AB	4.8	2 %	27.3	567 %	62.7	2.3	72 %	2.9	126 %	66.0
SK	0.6	0 %	3.6	569 %	8.9	0.0	78 %	0.3	1093 %	16.4
MB	0.3	2 %	1.5	600 %	3.5	0.0	91 %	0.1	n/a	8.2
ON	4.9	1 %	6.8	140 %	19.3	0.5	41 %	2.7	564 %	26.5
QC	3.9	34 %	0.5	13 %	4.1	0.0	42 %	3.2	n/a	21.0
NB	0.4	100 %	0.0	0 %	0.0	0.0	26 %	0.7	n/a	3.7
PE	0.2	2 %	0.6	285 %	2.1	0.0	74 %	0.1	n/a	1.4
NS	0.6	23 %	0.4	66 %	2.7	0.0	4 %	1.1	n/a	2.6
NL	0.1	4 %	0.7	1315 %	1.3	0.0	24 %	0.8	n/a	3.9
TOTAL	16.5		45.8		154.0	2.8		37.1		221.8

within each Canadian province. Existing installed capacities reflect current installations of utility-scale wind and solar generation and exclude planned additions. Wind generation currently operates in all provinces, while most provinces do not currently have installed utility-scale solar generation. Scenarios with no additional capacity reflect the proportion of modeled scenarios in which no wind or solar is developed to 2050. Mean and maximum additional installed capacity provide insights into the magnitude of the potential buildout of these technologies to 2050 within each province.

Findings indicate that six provinces (AB, MB, ON, PE, NL, SK) develop wind with high robustness, as reflected by the low proportion of scenarios having no additional capacity by 2050. In three provinces (BC, QC and NS) the extent of wind development depends strongly on inputs. Only one province (NB) lacks additional wind development due to anticipated low demand growth (Canada Energy Regulator, 2023b). Solar development occurs in all provinces but varies substantially by proportion of scenarios with no utility-scale solar development (4–91 %), by mean (0.1–25.2 GW) and by maximum (1.4–72.2 GW) developed capacity. Provinces developing significant wind (solar) resources are less likely to also develop significant solar (wind) resources.

3.1.1. British Columbia and Alberta

In BC, wind energy development exhibits high robustness, displaying limited sensitivity to input variations, as more than half of the ~2000 scenarios develop no new wind (Fig. 3). The y-axis on this histogram presents the count of scenarios returning technology deployment values within each bin across the maximum range of values. The x-axis contains normalized values in standard deviations from the mean, with negative values indicating scenarios where values fall below the average. Specifically, mean wind deployment in BC to 2050 totals ~4.3 GW (4344 MW), ranging between 0 GW in ~1100 scenarios up to ~49.4 GW (6 s.d.) in the most extreme scenario. This insensitivity to inputs implies a relatively stable path of limited future wind energy development in BC, a finding further explored in the Discussion.

Contrastingly, solar energy development in BC is characterized by greater variability and uncertainty that is significantly influenced by policies and input parameters, including the carbon tax, future relative capital costs of wind and solar, and electricity demand in both BC and Alberta (Fig. 3). The derivatives plot illustrates the effect of each of six factors on solar generation in BC to 2050. The maximum absolute derivative value indicates the steepest slope or the maximum change of one variable in relation to the other. Higher carbon taxes, higher wind capital costs and lower solar capital costs significantly increase development of solar generation, with demand in either province and the

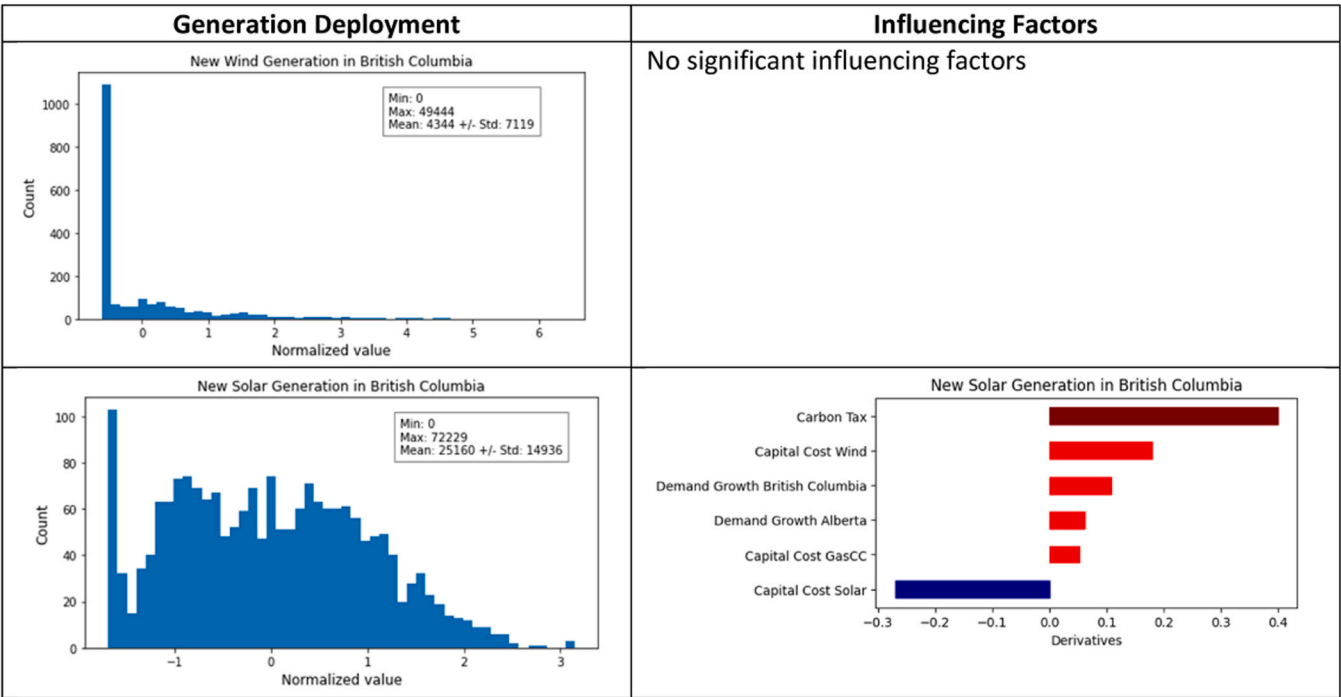


Fig. 3. Wind and Solar Generation in BC – Deployment and Influencing Factors.

capital cost of natural gas combined cycle generation having a more modest impact.

Wind energy is likely to play a substantial role in Alberta’s energy future regardless of evolving circumstances. Substantial new wind (mean ~27 GW) is developed to 2050, with larger capacity buildouts under a significant minority of scenarios (Fig. 4). The carbon tax is the primary factor influencing wind development, followed by lower wind capital costs and higher solar capital costs. Despite variations in inputs,

solar development in Alberta reflects lower sensitivity to input changes (and lower overall development) compared to wind energy (Fig. 4).

3.1.2. Saskatchewan and Manitoba

There are wide variations in the quantities of wind developed in Saskatchewan and Manitoba (Fig. 5). In almost all scenarios (> 95 %), new wind generation capacity is developed in Saskatchewan (>1000 MW) and in Manitoba (> 500 MW) by 2050. In rare scenarios

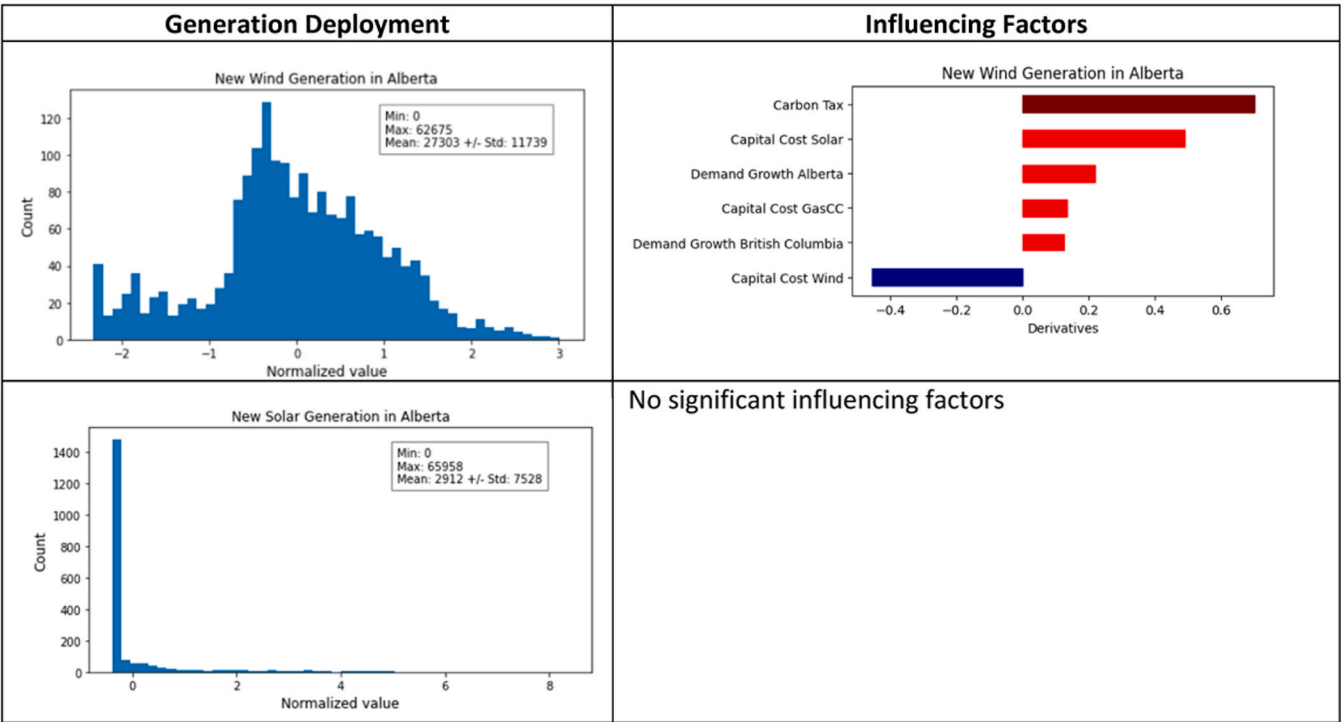


Fig. 4. Wind Generation in Alberta - Deployment and Factors.

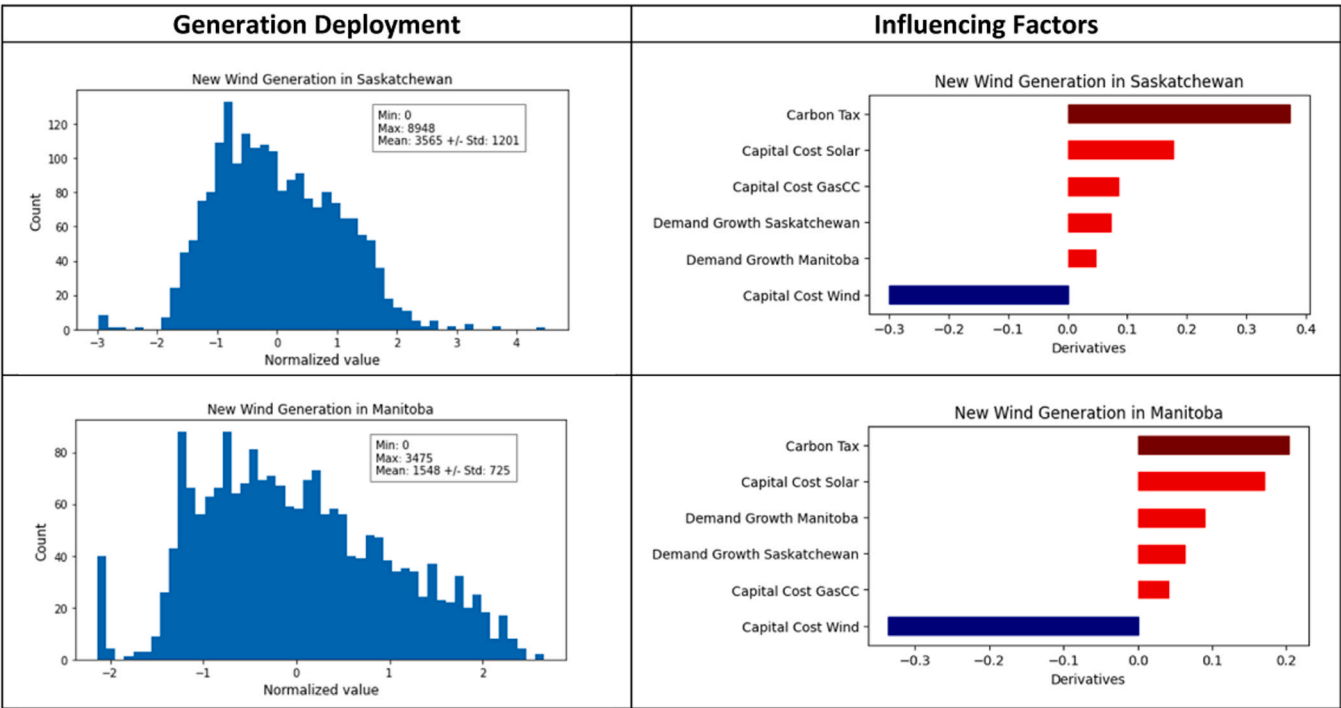


Fig. 5. Wind Generation in Saskatchewan and Manitoba – Deployment and Factors.

consisting of low carbon prices, low demand and high technology costs is no wind developed in either province. No new solar development occurs in most scenarios in Saskatchewan (>75 %) and in Manitoba (>90 %) suggesting a robust pathway of limited development to 2050 (Table 4). Mean solar development is relatively low in Saskatchewan (~300 MW) and lower still in Manitoba (~150 MW), while additional development occurs where future solar capital costs decline compared to

wind capital costs (Fig. 5).

3.1.3. Ontario and Quebec

Regardless of future conditions, wind energy will be a growing feature of Ontario’s electricity system, influenced by increasing carbon prices, and relative capital cost differences between wind and solar generation. Mean additional development by 2050 is substantial

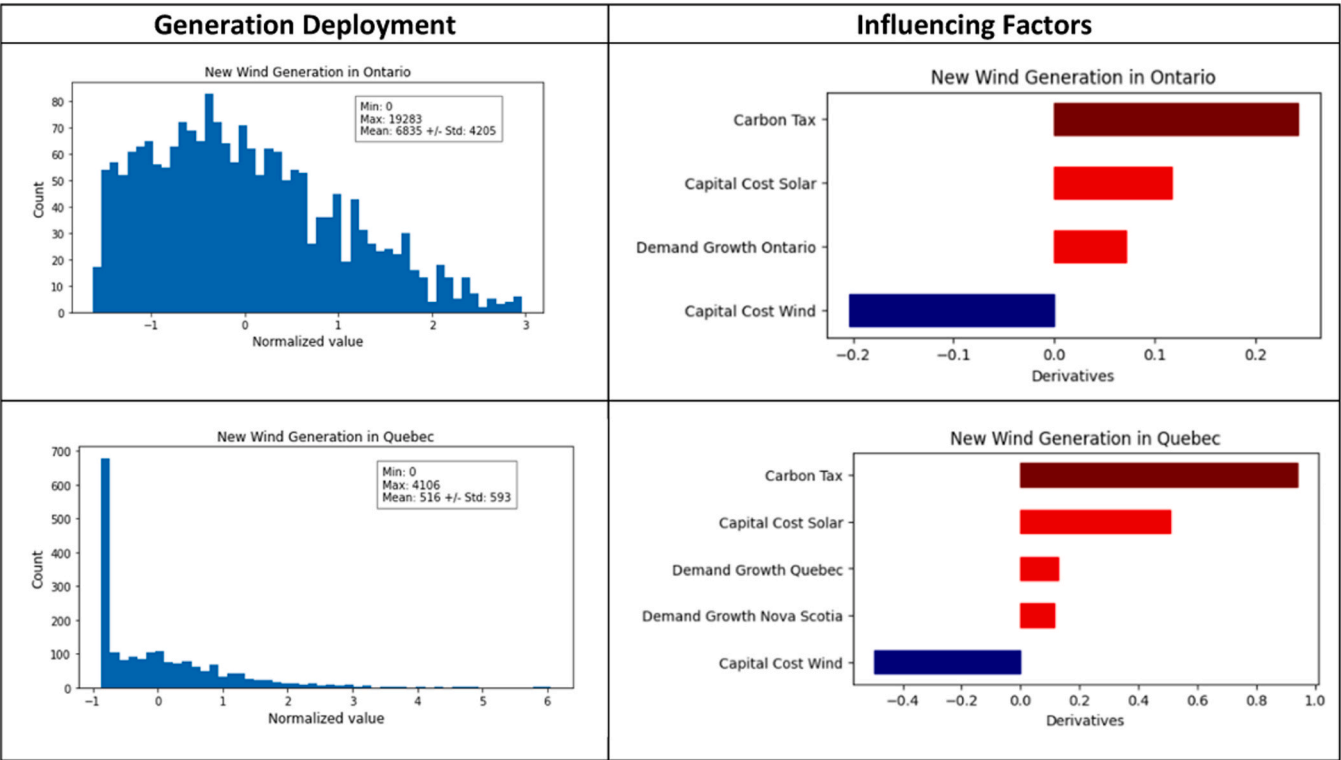


Fig. 6. Wind and Solar Generation in Ontario and Quebec – Deployment and Factors.

(~7 GW), and only in rare scenarios of high wind costs, low solar costs, low demand, and low carbon prices is no new wind energy developed in Ontario (Fig. 6). In a majority of scenarios (~60 %), new solar generation is developed in Ontario with mean development (~2.7 GW) indicating a somewhat less robust path compared to wind development.

In nearly two-thirds of scenarios, additional wind is developed in Quebec, though compared to existing installed capacity (~4 GW) mean development (~0.5 GW) is modest. As in Ontario, in a majority of scenarios (~60 %) new solar generation is developed in Quebec, though unlike in Ontario mean solar development (3.2 GW) exceeds mean wind development. The lack of development scenarios including either wind (34 %) or solar (42 %) in Quebec illustrates significant dependencies of wind and solar build-out on input conditions. Future wind generation deployment in Quebec is driven by increasing carbon prices, the relative capital costs of wind and solar energy, and to a lesser degree by demand in Quebec and in nearby provinces, while no single factor influences future solar deployment in Quebec or Ontario.

3.2. Natural gas generation

Findings for natural gas generation indicate pathways to deep decarbonization in all provinces do not retire these generating assets almost regardless of input variables (Table 5). This result is facilitated by

the model's lack of a hard constraint on emissions, which decline to net-zero but not to zero by 2050. Natural gas additions are limited to combined cycle generation, which has lower modeled capital and operating costs compared to simple cycle generation. For all provinces, with the exception of BC, natural gas additions are minimal with substantial additions occurring only under very extreme circumstances. This flexible thermal generation, along with hydroelectric generation across the country, supports integration of increasing quantities of variable renewable generation while providing contingency and reserve capacity. This indicates a robust pathway for existing installed natural gas generation to support decarbonization into the foreseeable future.

In BC, in contrast to other provinces, there is significant (mean 4.3 GW) expansion in new natural gas combined cycle generation. This consistent selection of new natural gas generation reflects its lower costs compared to other forms of dependable and flexible capacity, including hydroelectric generation and energy storage, and that nuclear generation is prohibited in BC (Clean Energy Act, 2010). Existing combined cycle generation under all scenarios and simple cycle under most scenarios (95 %) remains operational until 2050 (Fig. 7).

3.3. Interprovincial transmission

For all three provincial pairings investigated, a majority of scenarios

Table 5
Modeled natural gas development by Province.

NATURAL GAS COMBINED CYCLE							
Province	Existing	Retirements			Additions		
	Installed capacity	Scenarios with no capacity retirement	Mean capacity retired	Maximum capacity retired	Scenarios with no capacity additions	Mean additional installed capacity	Maximum additional installed capacity
	(GW)	(%)	(GW)	(GW)	(%)	(GW)	(GW)
BC	0.2	100%	0.0	0.0	18%	4.3	15.6
AB	13.5	55%	1.3	9.4	90%	0.3	10.7
SK	1.5	84%	0.0	1.1	95%	0.0	2.3
MB	0.0	100%	0.0	0.0	95%	0.0	1.3
ON	3.5	78%	0.1	0.5	89%	0.1	5.9
QC	1.0	55%	0.3	1.0	100%	0.0	0.0
NB	0.4	98%	0.0	0.1	91%	0.0	1.1
PE							
NS	0.0	87%	0.0	0.3	81%	0.0	0.4
NL							
NATURAL GAS SIMPLE CYCLE							
Province	Existing	Retirements			Additions		
	Installed capacity	Scenarios with no capacity retirement	Mean capacity retired	Maximum capacity retired	Scenarios with no capacity additions	Mean additional installed capacity	Maximum additional installed capacity
	(GW)	(%)	(GW)	(GW)	(%)	(GW)	(GW)
BC	0.0	95%	0.0	0.0	100%	0.0	0.0
AB	1.3	95%	0.0	1.1	100%	0.0	0.0
SK	0.6	95%	0.0	0.7	100%	0.0	0.0
MB	0.3	95%	0.0	0.4	100%	0.0	0.0
ON	6.9	100%	0.0	0.0	100%	0.0	0.0
QC	0.0	100%	0.0	0.0	100%	0.0	0.0
NB	0.0	100%	0.0	0.0	100%	0.0	0.0
PE							
NS	0.1	100%	0.0	0.0	100%	0.0	0.0
NL							

Not applicable as generation type does not exist in this jurisdiction

Note: Gray cells indicate the generation type does not exist in this province.

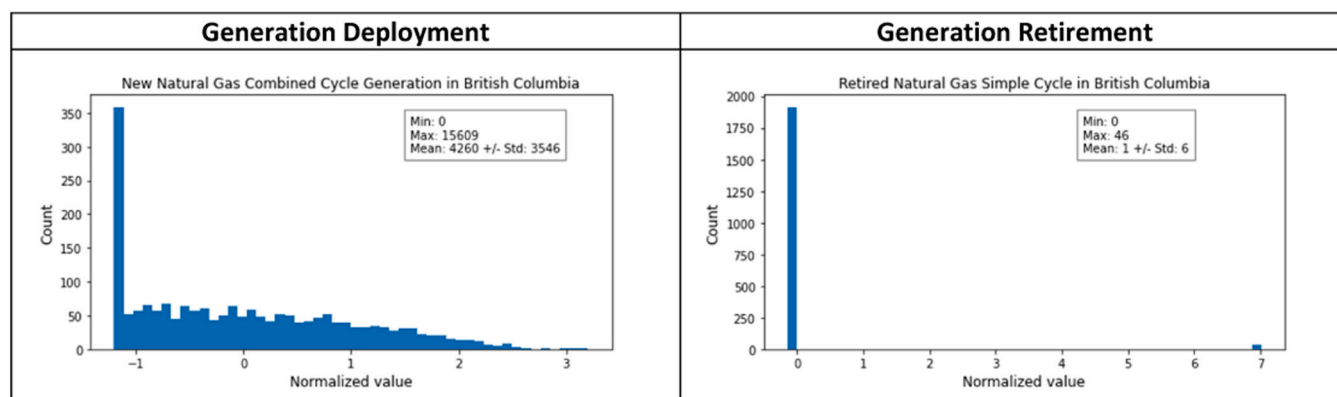


Fig. 7. Natural Gas Generation in BC – Deployment and Retirement.

see additional transmission transfer capacity between jurisdictions, suggesting an important role for transmission expansion in grid decarbonization in Canada (Table 6).

In the majority of scenarios, additional transmission transfer capacity is developed between BC-Alberta (~60 %), Saskatchewan-Manitoba (~85 %), and Ontario-Quebec (~75 %), with modeled development varying under different input conditions. Influencing factors across provincial pairings include higher wind costs reducing transmission development, and higher carbon taxes, higher solar costs and, to a lesser degree, demand growth increasing development. A high carbon tax results in more rapid decarbonization in provinces with comparatively higher emissions intensities (AB, SK, ON), triggering more interconnection with highly decarbonized neighbouring provinces (BC, MB, QC). Lower wind generation capital costs increase wind capacity, which results in larger surplus generation best utilized in neighbouring jurisdictions by developing additional interprovincial transmission. As solar capital costs decrease, the installed transmission between provinces decreases, an intriguing result explored further in the Discussion.

4. Discussion

The principal purpose of this investigation is to provide insights for policymakers into least-cost, robust pathways to net-zero within Canadian provincial electricity systems. Specific policy implications discussed below include those related to the federal *Greenhouse Gas Pollution Pricing Act*, *Clean Investment Tax Credit* and *Clean Electricity Regulations*, the recently announced Alberta renewables policy and the *BC Clean Electricity Delivery Standard*. A secondary interest is to investigate the potential of ML models to analyze least-cost optimization of power systems to net-zero emissions in response to uncertainties and variability in technology capital costs, electricity demand, and carbon pricing.

4.1. Renewable generation

Our findings indicate that four provinces need to plan for significant new wind development to 2050, including Alberta (mean 27.3 GW, max. 62.7 GW), Ontario (mean 6.8 GW, max. 19.3 GW), Saskatchewan (mean 3.6 GW, max 8.9 GW) and Manitoba (mean 1.5 GW, max 3.5 GW), with other provinces also likely to see additional wind development (Table 1). Solar development occurs in all provinces but varies substantially in terms of developed capacity. The primary driver of deployment is the federal carbon tax, indicating that increasing the price on carbon beyond the 2030 date currently mandated in the federal *Greenhouse Gas Pollution Pricing Act* is indispensable to deep decarbonization. Technology capital costs also play a prominent role in driving renewables deployment. The recently proposed *Federal Clean Electricity Investment Tax Credit* introduces a 15 % refundable tax credit for non-emitting electricity generation, including wind, solar and abated natural gas generation, as well as interprovincial transmission (Chapter 3: A *Made-In-Canada Plan*, 2024). To support investment, this investment tax credit needs to be predictable while also responsive to the potential for periodic stagnation and for rapid declines in the capital costs of renewables, both of which characterize the historical costs trends for wind (Lantz et al., 2024) and solar generation (Solar Installed System Cost Analysis, 2024).

At the provincial level, Alberta has high suitability for both wind and solar generation, most (72 %) of the investigated scenarios suggest no additional solar capacity to 2050, and modest mean future solar development (~3 GW) compared to wind (~27 GW) (Fig. 3). This can be explained by the differing contribution of wind and solar generation to reserve requirements, which are more challenging to meet in a decarbonized Alberta grid given its limited supply of hydroelectric generation (Fig. 2). The estimated year-round effective load carrying capability (ELCC) of wind generation (16 %) in Alberta (Annual Market Statistics Reports » AESO, 2023) makes a meaningful dependable capacity contribution to reserve requirements reducing the need for natural gas capacity. The estimated ELCC of solar generation in Alberta in summer

Table 6

Modeled transmission transfer additions by provincial pairs.

TRANSMISSION						
Province	Province	Existing		Additions		
		→ transfer capacity	← transfer capacity	Scenarios with no additional capacity	Mean additional transfer capacity	Maximum additional transfer capacity
		(GW)	(GW)	(%)	(GW)	(GW)
BC	AB	1.00	0.80	42 %	1.50	9.00
SK	MB	0.15	0.40	17 %	0.22	1.03
ON	QC	2.04	2.67	27 %	1.05	4.95

Note: While existing transfer capacities are presented unidirectionally, the modeled transmission additions are bi-directional.

(20 %) and winter (0 %) means that solar makes no winter contribution to reserve requirements in Alberta, a winter-peaking system ([Annual Market Statistics Reports » AESO, 2023; AESO, 2024](#)). This reserve contribution difference tilts the preference toward wind over solar generation in Alberta. Nationally, the ELCC of wind generation is typically 15–35 % of installed capacity with higher contributions in winter than summer ([BC Hydro, 2023; Annual Market Statistics Reports » AESO, 2023; IESO, 2023b](#)). Across Canadian provinces, solar generation makes no contribution to winter reserve requirements, though almost all provincial systems are winter peaking ([BC Hydro, 2023; Annual Market Statistics Reports » AESO, 2023; IESO, 2023b](#)). Wind generation thus has a larger impact compared to solar generation in reducing the need for higher-emitting load-balancing generation. Cost is not the sole factor determining whether to proceed with wind or solar generation as operational considerations are also relevant ([Government of Alberta, 2023](#)).

Alberta recently placed a seven-month pause on utility board approvals related to wind and solar facilities to conduct an inquiry into environmental impacts, site reclamation, and electricity system reliability ([Alberta Utilities Commission, 2024](#)). In response to inquiry findings, the Government of Alberta announced a prohibition on wind and solar development on Class 1 and 2 agricultural lands and on wind development within a 35-km buffer zone surrounding existing and future protected areas ([Government of Alberta, 2024](#)). This recent policy direction would preclude wind, and to a lesser extent solar, development over large areas of Alberta not considered in the current analysis. Additional investigation should assess the implications of this nascent policy, which would make unavailable some lower cost locations for wind and solar development. As a result, a higher carbon price would be required to shift deployment towards renewables and away from natural gas, delaying decarbonization. The sum of modeled mean wind deployment across all provinces in the current analysis (~46 GW) vastly exceeds total existing installed capacity (~16 GW) ([Table 4](#)). Realizing this deployment requires policy alignment across federal, provincial, and municipal governments incorporating best practices for collaborative development that considers rural, agricultural, human health, species protection and wilderness values ([Jeyakumar and Todd, 2024](#)). Neither COPPER nor the current ML model account for these landscape variables that could preclude wind development. Future modeling could utilize the current ML model to test variations in land availability (e.g., parklands, agricultural lands, waterbodies, etc.) and consequent effects on generation technology deployment and system costs.

In BC, solar generation may be poised for significant expansion under appropriate cost and policy conditions. Modeled mean solar development (~25 GW) greatly exceeds modeled mean wind development (~4 GW), despite wind generation currently being the lowest-cost source of low-carbon energy in the province ([BC Hydro, 2023](#)). Two factors potentially contribute to this result. First, requirements for additional generation in BC are delayed due to the Site C Project, a large (1100 MW, 5200 GWh/y) hydroelectric facility that enters the COPPER model in 2025. Since our modelling inputs include greater declines over time in the capital costs of solar generation compared to wind generation, this delay results in greater proportional solar deployment into the future. Additionally, prime solar development locations in BC are relatively proximal to the existing transmission network, resulting in comparatively lower transmission interconnection costs.

4.2. Fossil generation

Provincial electricity system operators plan for significant emissions reductions by taking different approaches to retirements, operations, and renewals of existing fossil-fueled generation. The Ontario Independent Electricity System Operator determined that natural gas generation provides dependable capacity, quick response, ramping capability, and reliability services while contributing to reserve requirements ([IESO, 2023a](#)). This combination of characteristics is

difficult to replicate, as comparable low-carbon generation resources are expensive and require substantial lead times, while storage and demand response would be required in unattainable quantities, leaving natural gas generation in operation beyond 2035 ([IESO, 2023a; IESO, Natural Gas Phase-Out Study, 2021](#)). Furthermore, a rapid phase-out of natural gas generation in Ontario would result in frequent blackouts ([IESO, Natural Gas Phase-Out Study, 2021](#)). Similarly, the Alberta Electricity System Operator found that pathways to net-zero by 2035 were highly uncertain, costly, and required verified emission offsets and widespread application of unproven technologies ([AESO, 2022](#)). For its part, Hydro Québec is planning to retain and gradually convert remaining natural gas generation to operate on non-emitting fuels so that operational benefits and contributions to reserve requirements can be sustained ([Hydro Québec, 2023](#)).

Contrastingly, BC Hydro is implementing a 100 % Clean Electricity Delivery Standard, which would see the phasing out of remaining natural gas generation by 2030 ([Government of BC, 2022](#)), a strategy potentially at odds with the current study's findings. While most modeled scenarios across all provinces (>90 %) see no new natural gas development ([Table 5](#)), the current analysis indicates that least-cost decarbonization pathways develop significant natural gas generation (mean ~4.3 GW) in BC to 2050. This finding is consistent with a recent assessment of integrated continental electricity decarbonization, which found that natural gas generation continues to be developed to 2050 within BC and Alberta ([Brinkman et al., 2024; NREL, 2021](#)). A more recent study examining options for 100 % clean electricity in the US by 2035 found that unabated natural gas combustion turbines provide a significant proportion (21 %) of system dependable capacity while generating a small proportion (4 %) of annual energy ([NREL, 2022](#)). Though emissions from unabated generation can potentially be offset, there is a high-cost premium to decarbonizing the last portion of electricity sector emissions ([NREL, 2022](#)). These collective findings suggest that advancing the 100 % Clean Electricity Delivery Standard in BC, which would decommission remaining natural gas generation by 2030 while prohibiting new unabated generation ([Government of BC, 2022](#)), should be preceded by thorough consideration of alternatives, including retaining and developing natural gas generation while investigating low-carbon fuels ([Canada Energy Regulator, 2023a](#)) and carbon offsets ([AESO, 2022](#)), to mitigate the effects of rapid grid decarbonization on system-wide costs.

The current analysis models to the policy context in place at the time of modelling, while policy is rapidly evolving in response to decarbonization imperatives. In August 2023, the Canadian government tabled Draft Clean Electricity Regulations (CER) which apply a revised emissions standard (30 tCO₂e/GWh) to interconnected fossil-fueled generation beginning on January 1, 2035, or 20 years following facility commissioning, whichever is later ([Government of Canada, 2023b](#)). More recently, Environment Canada suggested increasing the emissions standard to permit abated natural gas generation (i.e., with carbon capture and storage) that operates primarily as load-following or peaking generation, and therefore cannot meet such a restrictive standard ([Government of Canada, 2023b](#)). Consistent with these contemplated revisions, the current analysis illustrates robust pathways to deep decarbonization in which almost all existing natural gas generation remains on-line, operating less frequently as the system decarbonizes. In this analysis, the carbon tax is the most influential factor affecting generation and transmission development and acts as the primary constraint on the retirement and deployment of natural gas generation. In terms of retirements, following 2035 the lower performance standard contemplated in the Draft CER would further reduce annual capacity factors at many existing facilities, potentially triggering retirements earlier than forecast in the current analysis. Considering the low capital cost of natural gas generating units, the Draft CER may also have the effect of triggering retrofits of existing facilities to operate on non-emitting fuels to achieve emissions reduction targets while maintaining reliability. In terms of deployments, the current analysis does not

project significant deployments other than in BC, so adoption of the Draft CER would not substantially alter results for natural gas deployments.

4.3. Transmission

Results for interprovincial transmission development provide insights into the complex interactions between carbon taxes, generation technology costs and electricity demand. Findings for the three provincial pairings investigated indicate that lower wind generation capital costs increase transmission development between provinces, while lower solar generation costs decrease development. Lower wind capital costs result in more wind deployment and larger daily and seasonal surplus generation, which is managed through exports to neighbouring jurisdictions facilitated by additional interprovincial transmission development. By utilizing the geographic variability in the wind resource across two provinces, transmission expansion enables exports during wind energy surpluses and imports during shortfalls. Contrastingly, lower solar capital costs trigger more solar deployment and larger intra-daily surpluses most cost-effectively managed through shifting of load-following generation rather than by developing additional interprovincial transmission.

Increasing electricity demand also influences transmission development, though to a lesser extent than carbon taxes and wind capital costs. In general, demand in the province in which less wind development is

projected has a greater influence on interprovincial transmission development. In other words, increasing demand in BC, MB, and QC influences interprovincial transmission development more than does increasing demand in AB, SK or ON. This result is unsurprising considering that wind generation creates surpluses best utilized by neighbouring provinces but only if their demand justifies interprovincial transmission development.

In the case of Ontario-Quebec transmission, solar capital costs influence the need for new transmission more than do higher carbon taxes, a unique result. Higher solar capital costs also increase wind development in Quebec (Fig. 6). As solar capital costs increase, it is preferable to incur the costs of additional transmission in order to trade more electricity between these two provinces. In the case of BC-Alberta, the factor analysis suggests a similar mechanism driving transmission development as between Ontario and Quebec (Fig. 8). The additional findings that wind development is preferred in Alberta and solar development in BC suggest that given lower wind capital costs, Alberta develops large quantities of wind along with transmission to support exports to BC. This is supported by the additional influence of demand growth in BC on transmission development between the provinces (Fig. 8). However, given lower solar capital costs, the provinces each develop their own domestic resources, mainly solar in BC and wind in Alberta, without expanding interprovincial transmission.

These somewhat complex interactions between solar costs, wind costs and interprovincial transmission development have implications

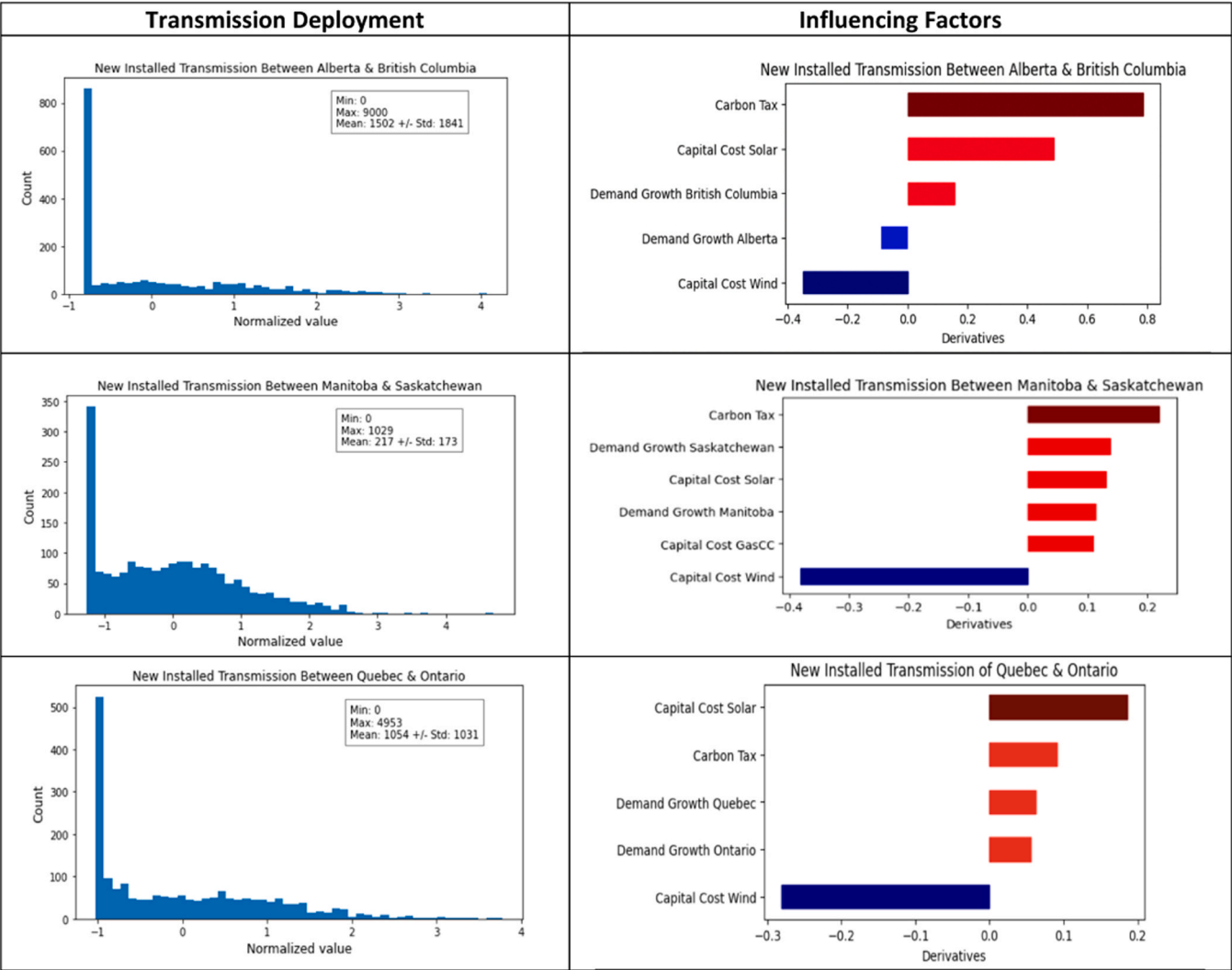


Fig. 8. Transmission between provinces – Deployment and Factors.

for policymaking, particularly for funding allocations under the *Clean Investment Tax Credit*. Specifically, a marked decline in the future cost of electricity from solar compared to wind generation suggests a more limited requirement for investment in interprovincial transmission and increased investment in technologies (e.g., energy storage, abated natural gas generation) for balancing diurnal solar variations. Contrastingly, a marked decline in the costs of electricity from wind relative to solar generation indicates a need for greater investment in interprovincial transmission. Incentives and investments in interprovincial transmission should be strategically considered in terms of the evolution of wind and solar generation technology costs.

5. Conclusions

This research investigates the potential of ML models to analyze least-cost decarbonization of provincial power systems across Canada in response to uncertainties and variability in technology capital costs, electricity demand growth, and carbon pricing. The study provides several insights into robust pathways respecting wind and solar generation, reducing emissions from natural gas generation, and expanding Canadian interprovincial transmission that remain robust under many uncertainties and variabilities.

Findings indicate robust conditions for substantially expanded wind generation but less robust conditions for expanded solar generation in the decarbonization of Canada's provincial electricity grids. Most provinces develop wind under almost all future scenarios, with the potential exception of British Columbia, where higher-cost wind generation results in solar generation becoming the primary low-carbon electrical energy source.

While Canadian policy is increasingly directed towards decarbonizing electricity grids, our analysis indicates that existing natural gas generation is maintained but infrequently operated in almost all scenarios in all provinces, providing support for policies maintaining these resources for reliability and reserve requirements. Almost no new natural gas is developed under any scenarios, with the noted exception of British Columbia. Carbon prices, along with wind and solar capital costs, act as primary determinants of least-cost decarbonization pathways. Considerations for planners and policymakers involve optimizing the use of existing natural gas generation while also strategizing gradual phase-out of emissions in line with decarbonization goals and system reliability.

In terms of interprovincial transmission, our findings indicate that provinces should proactively and strategically explore opportunities to enhance transmission transfer capabilities, ensuring efficient energy exchange to meet regional demands. This exploration, including potential federal investment, should be cognizant of the influence of lower wind generation costs in increasing the development of interprovincial transmission as part of a least-cost pathway to decarbonization, and lower solar generation costs decreasing the need for interprovincial transmission.

Recent analyses of pathways to deep decarbonization reveal significant future wind and solar development across Canada (Arjmand and McPherson, 2022; IESO, 2023a; AESO, 2022) demonstrate that the extent of development within individual provinces is the outcome of factors considered in the current analysis, including future demand growth, relative technology costs, and decarbonization policies, including carbon prices. However, other potential factors not considered here include variations in the quality of the wind resource (i.e., wind speed) or solar resource (i.e., insolation), the distance from these resources to load centers, the influence of landscape characteristics, and the costs of interprovincial transmission. Resource quality and distance to grid are inputs to COPPER and considered to be accurately quantified with limited potential for variation. Future modeling could further test the influence of variations in land availability and interprovincial transmission capital costs on grid decarbonization in Canadian provinces.

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CRediT authorship contribution statement

Madeleine McPherson: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Richard Hendriks:** Writing – review & editing, Writing – original draft, Conceptualization. **Zahra Jahangiri:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data is now available and public and the links are provided.

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